

主定理证明

$$T(n) = aT(\frac{n}{b}) + f(n) \quad \text{比较 } f(n) \text{ 与 } n^{\log_b a}$$

$$\text{case 1: } f(n) = O(n^{\log_b a - \epsilon}) \quad (\epsilon > 0), \text{ 则 } T(n) = \Theta(n^{\log_b a})$$

$$\text{case 2: } f(n) = \Theta(n^{\log_b a}), \text{ 则 } T(n) = \Theta(n^{\log_b a} \lg n)$$

$$\text{case 3: } f(n) = \Omega(n^{\log_b a + \epsilon}), \text{ 且 } af(\frac{n}{b}) \leq cf(n), \text{ 则 } T(n) = \Theta(f(n))$$

$(\epsilon > 0) \qquad (c < 1)$

$$\text{令 } n = b^k$$

$$\text{则 } T(n) = a(aT(\frac{n}{b^2}) + f(\frac{n}{b})) + f(n)$$

$$= a^k T(1) + \sum_{m=0}^{k-1} a^m f(\frac{n}{b^m})$$

$$= n^{\log_b a} + \sum_{m=0}^{k-1} a^m f(\frac{n}{b^m})$$

↓

所有最小子问题的计算工作量

↳ 规约到子问题及综合解的工作量

$$\therefore \text{对于 case 1: } T(n) = n^{\log_b a} + O(\sum_{m=0}^{k-1} a^m (\frac{n}{b^m})^{\log_b a - \epsilon})$$

$$= n^{\log_b a} + O(n^{\log_b a - \epsilon} \sum_{m=0}^{k-1} (\frac{a}{b^{\log_b a - \epsilon}})^m)$$

$$= n^{\log_b a} + O(n^{\log_b a - \epsilon} \cdot \frac{1 - b^{\epsilon \log_b n}}{1 - b^{\epsilon}})$$

$$= n^{\log_b a} + O(n^{\log_b a - \epsilon} n^{\epsilon})$$

$$= \Theta(n^{\log_b a})$$

$$\text{对于 case 2: } T(n) = n^{\log_b a} + \Theta(\sum_{m=0}^{k-1} a^m (\frac{n}{b^m})^{\log_b a})$$

$$= n^{\log_b a} + \Theta(n^{\log_b a} \sum_{m=0}^{k-1} 1)$$

$$= n^{\log_b a} + \Theta(n^{\log_b a} \cdot \log_b n) = \Theta(n^{\log_b a} \lg n)$$



对于 case 3: $T(n) = n^{\log_b a} + \sum_{m=0}^{k-1} a^m f\left(\frac{n}{b^m}\right)$

$$\leq n^{\log_b a} + \sum_{m=0}^{k-1} c^m f(n)$$

$$= n^{\log_b a} + f(n) \cdot \frac{1 - c^{\log_b n}}{1 - c}$$

$$= n^{\log_b a} + O(f(n))$$

$$= O(f(n))$$

